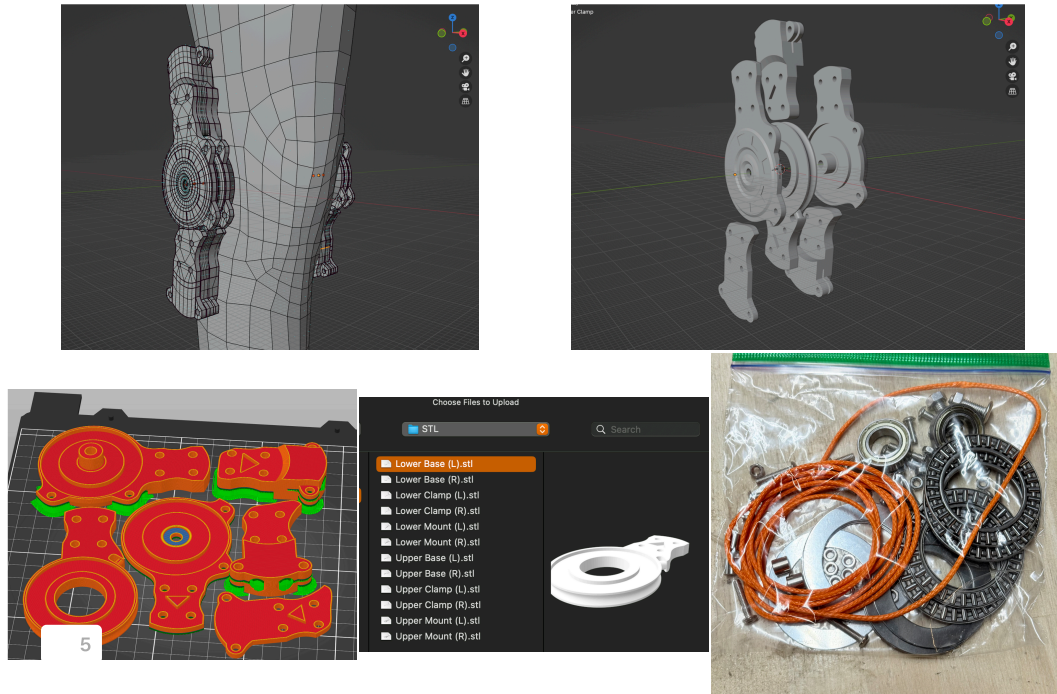




Knee Exo Hinge

 **Crash Makerspace**
28 sept 2025

 6  38  

These are the STL files for printing the Cabled Knee Hinge with tension cable track, over-extension, and threaded rod interfacing!

PRINTING

I recommend dense gyroid infill (40%), and at least 5 perimeter shells to ensure that the rod mounts are solid. I've included an image of the recommended print orientation for each hinge. I printed these on a MK4 Prusa machine with 0.4mm nozzle.

HARDWARE

I sourced batches of various hardware off many vendors across A (a ton of hardware all at once to pick at iterate from), thus, I have th of the bearings in use. You can get a Hardware Pack from me (limi by sending \$55 (includes \$10 shipping and is U.S. only) via PayPal include your address. This will include the hardware below (exclud for 2 hinges and includes shipping, which is likely your cheapest c down from the bulk batch to just what you need for two hinges - ir 100 of parts where you only need 2 of. Otherwise, these are the me and quantity (for two hinges):

- X4 59mm Roller Bearings

Exo Hinges

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- X2 [15x28x7 Bearing](#)
- X8 60mm Washer at 0.9mm
- Dyneema Cable of 1mm to 2mm
- X4 [M4 Bolts 12mm](#)
- X8 [M4 Bolts 18mm](#)
- X16 [M4 Bolts 20mm](#)
- X28 [M4 Nuts](#)
- X2 [M6 Axle Bolt 20mm](#)
- X2 [M6 Axle Nut](#)
- X4 [M4x8 Bushings](#)
- X2 [Threaded Rods](#) (I go with 3ft and cut them to length)
- X2 [Springs](#) (I got mine at a tractor hardware store, but the steel is about 2.5mm. You can go with something like the one linked. Due to spring steel elongation & circumference of hinge track, I would stay between 160mm to 200mm long for these)

ASSEMBLY 🛠️

For the threaded rod attachment, I find that a single layer of tightly wrapped electrical tape improves grip when tightening the bolts (more than this and they won't fit). For the cable, I recommend a 1mm high strength cable and tying it around the lower base's printed bridge to connect. Up to tensioners or motors up stream. These hinges are very similar to what was shown in my [Build Log 1](#) on YouTube, minus a few updates which I had for making them armored like an exterior cap / vented insets / etc.

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Archivos adjuntos

Descargar todo

- Lower Base (L).stl
- Upper Base (R).stl
- Upper Clamp (L) (Easy Print Version).stl
- Lower Base (R).stl
- Upper Mount (L).stl

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- Lower Mount (L).stl
- Upper Clamp (L).stl
- Upper Clamp (R) (Easy Print Version).stl
- Upper Base (L).stl
- Lower Clamp (R).stl
- Upper Clamp (R).stl
- Lower Clamp (L).stl
- Lower Mount (R).stl
- Upper Mount (R).stl

38 comentarios

Únete a la conversación...

Noah B · 7d.

Do you have any hardware packs available? If so, how long would the shipping take to St. Louis?

4

Crash Makerspace AUTOR · 7d.

I've got one more—mm, probably 3 to 4 days to arrive

[Cargar respuestas](#)

Wolfy · 3s.

Do you happen to have more of those hardware packs available? If not its all good

4

Crash Makerspace AUTOR · 3s.

I have 1 pack left! It replaces the 18mm bolts with 20mm' the 18's

[Cargar respuestas](#)

Dan Zimmerman · 2/mes

yo, if you've got any more of the hardware packs would love to pick one up.

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Dan Zimmerman · 2/mes ...

No worries if not tho.



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Kawika · 3/mes ...

Are those hardware packs still available? If not no worries at all!



Crash Makerspace AUTOR · 3/mes ...

Mmmm I'd have to double check tomorrow afternoon to see if I still have stock! I'll get back to you



[Cerrar respuestas](#)



Lucas Gulick · 4/mes ...

Are hardware packs still available?



Crash Makerspace AUTOR · 4/mes ...

Yep! I can gather those for you



[Cargar respuestas](#)



Ahsan Tariq · 5/mes ...

"X8 60mm Washer at 0.9mm" - I believe that is a typo, did you mean 16mm? 60 is a bit crazy for an M8 outer diameter washer.



Crash Makerspace AUTOR · 5/mes ...

The washer isn't for an M8 it's to go above and below the needle bearings



[Cerrar respuestas](#)



Bendix Sondermann · 5/mes

Also: The Part that holds the Threaded Rods, instantly Breaks at the Screws are placed. Even without any Kind of Load. Which makes the Lines that need to be filleted, at least 1mm.



Crash Makerspace AUTOR · 5/mes

Is this around the slits on the sides of the mount that are I



[Cargar respuestas](#)



Bendix Sondermann · 5/mes

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Some Information so you can Update this: the Roller Bearings are: "40x60x3mm AKX Series Roller Needle Bearing", easy to find on Ali by multiple Vendors...then the 15x28x7mm Bearings are called "Deep Groove Ball Bearing 6902".



Luke · 6/mes



Are you gonna post the spring cover?



Crash Makerspace AUTOR · 6/mes



I plan too, I still have an issue with it cracking at the end (not a big functional issue, but can cause it to vibrate more



[Cargar respuestas](#)



enmanuel cuevas · 6/mes

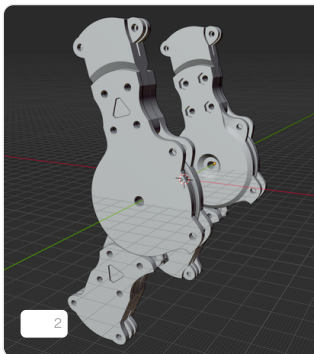


Hello, can you tell me which springs you use and where it is preferable to buy everything if I live in Latin America?



[Cargar más comentarios](#)

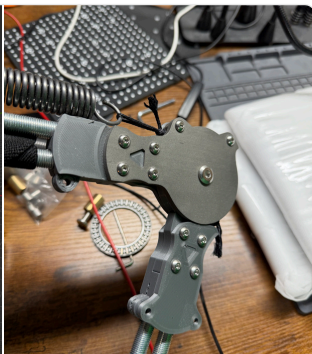
Publicaciones relacionadas >



Metal Exo Knee

These are the STL files for printing the Cabled Knee Hinge out of Stainless Steel or Titanium. I've adjusted the files for 3D Printing via S

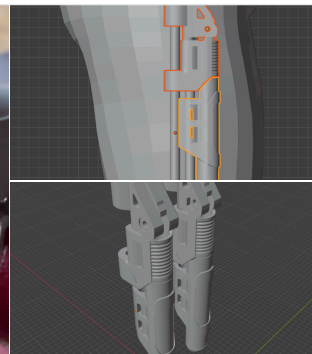
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Knee Exo Spring Mount & Guide

These are the files for the Spring's Guide-Cover and upper mounting block. Hardware is the same as listed in the base Exo Knee Hinge Post...

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Publicaciones populares >

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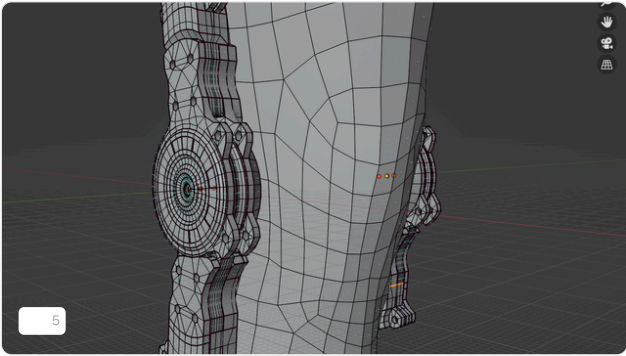


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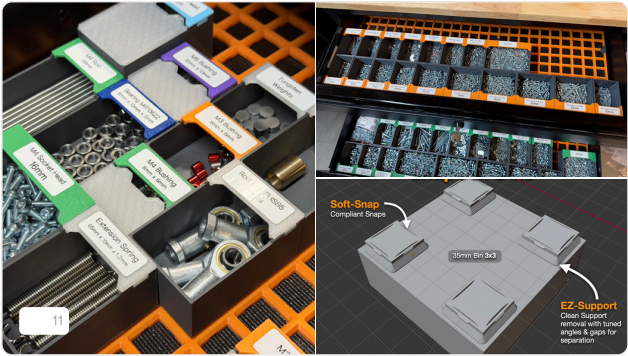




Knee Exo Hinge

These are the STL files for printing the Cabled Knee Hinge with tension cable track, over-extension, and threaded rod interfacing! PRINTING

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Built around a 25mm cell grid (significantly higher resolution than Gridfinity's 42mm), Snap-Cell Organizer enables tighter layouts,...

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